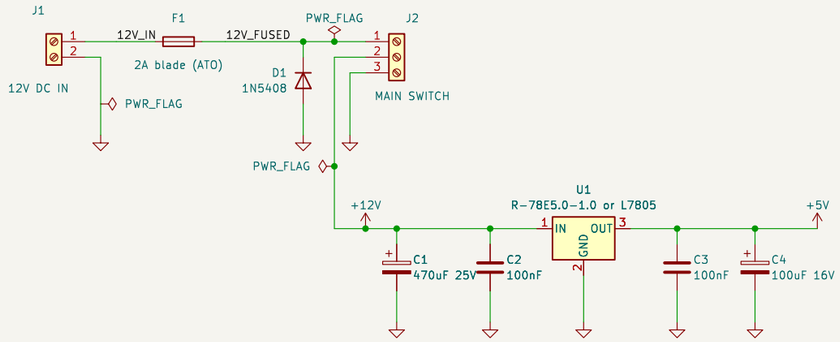
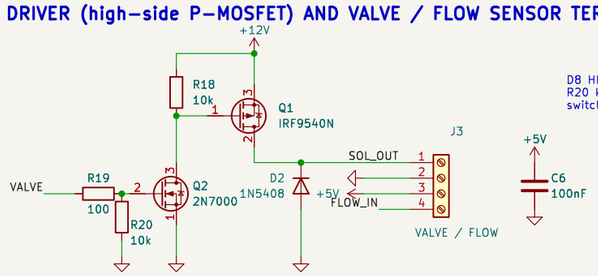
POWER IN: BLADE FUSE, MAIN SWITCH, REVERSE-POLARITY CROWBAR, 5 V REGULATOR



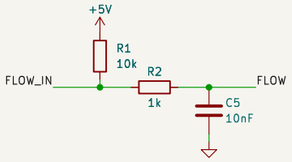
F1 + D1: if the supply is connected backwards, D1 conducts and blows F1 (the fuse comes before the main switch, so this works with the switch off too). J2 goes to the illuminated rocker switch; without a switch, link J2 pins 1 and 2. U1: the ESP32-C3 Wi-Fi peaks take the 5 V load to about 0.5 A. Use a pin-compatible switching regulator (R-78E5.0-1.0), or an L7805 on a heatsink.

VALVE DRIVER (high-side P-MOSFET) AND VALVE / FLOW SENSOR TERMINAL



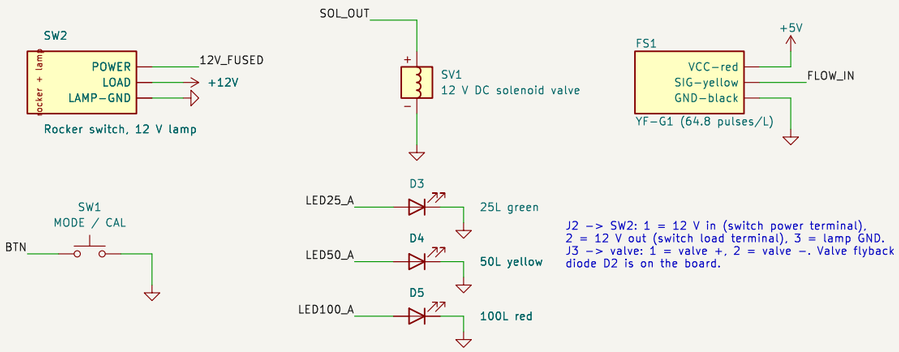
D8 HIGH: Q2 pulls the Q1 gate low and the valve gets 12 V. R20 keeps the valve off while the Nano starts. D2 clamps the switch-off spike right at the terminal.

FLOW SENSOR INPUT (YF-G1, NPN open collector)



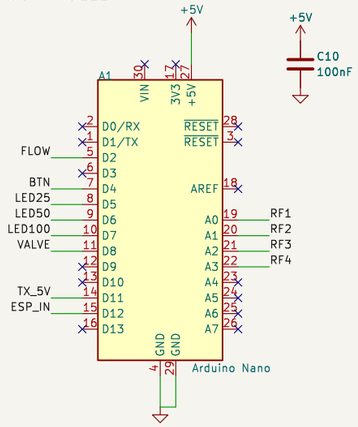
R2 + C5 filter fast spikes (time constant 10 us); the firmware also ignores pulses < 2 ms apart.

OFF-BOARD PARTS (wired to J2, J3, J5-J8; not placed on the PCB)



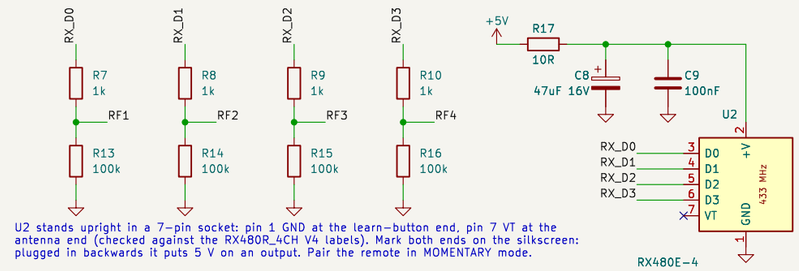
J2 -> SW2: 1 = 12 V in (switch power terminal), 2 = 12 V out (switch load terminal), 3 = lamp GND.
J3 -> valve: 1 = valve +, 2 = valve -. Valve flyback diode D2 is on the board.

CONTROLLER



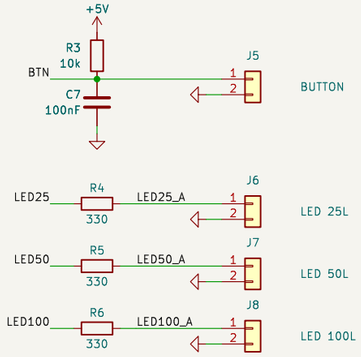
D11 sends status to the ESP; D12 receives from it (ESP control, for later). Nano is powered through its 5V pin (VIN unused).

REMOTE RECEIVER (RX480E 433 MHz module, plugs into the board)

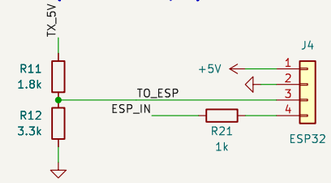


U2 stands upright in a 7-pin socket: pin 1 GND at the learn-button end, pin 7 VT at the antenna end (checked against the RX480R_4CH V4 labels). Mark both ends on the silkscreen: plugged in backwards it puts 5 V on an output. Pair the remote in MOMENTARY mode.

PANEL BUTTON AND LIMIT LEDS (JST-XH 2-pin connectors)



ESP32 LINK (JST-XH 4-pin)



MOUNTING HOLES (M4)



INTERFERENCE AND RELIABILITY MEASURES

1. D2 flyback diode at the valve output; Q1 switches the valve + (high side), valve - goes to GND.
2. 2 mm tracks for the 12 V and valve current; GND poured on both layers.
3. Decoupling: C1 + C2 on 12 V, C3 + C4 at U1, C10 at the Nano, C6 at J3.
4. Flow input: R1 10k pull-up, R2 1k + C5 10nF filter, firmware glitch filter.
5. Button: R3 10k + C7 100nF. Remote: R7-R10 1k series, R13-R16 100k pull-downs, receiver supply filtered by R17 + C8 + C9. Firmware debounces both.
6. F1 fuse (before the main switch) + D1 crowbar protect against shorts and a reversed supply.
7. R11 / R12 divide the 5 V data line to 3.2 V for the ESP; R21 protects D12 from the ESP.
8. Twist the flow sensor signal with its GND; route it away from the valve and 12 V wires.
9. Receiver antenna end at a board edge, away from the valve driver and the regulator.

Pins: D2 flow, D4 button, D5-D7 LEDs, D8 valve, D11/D12 ESP, A0-A3 remote
Firmware: WaterValveController.ino
YF-G1 flow sensor, MOSFET valve driver, RX480E 433 MHz receiver, ESP link
Arduino Nano carrier board

Sheet: /
File: WaterValveController.kicad_sch

Title: Water Valve Controller

Size: A3	Date: 2026-10-12	**Rev: 1.2**
KiCad E.D.A. kicad-cli 7.0.11+dfsg-1build4		Id: 1/1